Projects
Machine Learning Course
Fall 2026

EPFL
School of Computer and Communication Sciences
Nicolas Flammarion & Martin Jaggi
www.epfl.ch/labs/mlo/machine-learning-cs-433/
epfmlcourse@gmail.com

Class Project 2
Submission Deadline: Dec 17th, 2026

Introduction

In this project, you will learn to use the concepts we have seen in the lectures and practiced in the labs on a real-world dataset, start to finish. You will be doing exploratory data analysis to understand your dataset and your features, do feature processing and engineering to clean your dataset and extract more meaningful information, implement and use machine learning methods on real data, analyze your model and generate predictions using those methods and report your findings.

Grading. Project 2 counts 30% to your final grade in the course. This year, the project is done as a Machine Learning for Science project in collaboration with a research lab (see below).

Logistics

Group formation. For Project 2, you again form a team of 3 members on your own. It can be different from your group in Project 1. If you are still searching for teammates, please use the discussion forum. In exceptional cases, if you didn't manage to form a team of 3 by Nov 18th, contact us.

Submission deadline. Dec 17th, 2026 (at 16:00 afternoon, sharp)

Machine Learning for Science

Pick a real-world challenge offered by any research group of the (extended) EPFL campus, subject to availability. You learn about an interesting application domain, and collaborate with the lab to apply machine learning methods to their specific research question, on real data provided by the lab - an exciting option to follow an interdisciplinary approach to find new insights with your team.

https://www.epfl.ch/labs/mlo/ml4science/

Deliverables at a glance. (More details and grading criteria further down)

- **Written Report.** You will write a maximum 4 page PDF report on your findings, using LaTeX.
- **Code.** In Python. External libraries are allowed, if properly cited.

  The guidelines for the projects are explained below.

Participation of the Hosting Lab. You do this project in collaboration with another lab on EPFL campus. The lab hosting you will help us grading the domain-specific merit of your contribution.

We encourage you to reach out to any research group of the EPFL campus, or any academic institution in Switzerland (such as EPFL, UniL, CERN, CHUV, Idiap, and others). Labs interested in hosting a student group will contact us here to offer their project idea, dataset or task.

Here is a list of labs which already have proposed several exciting project ideas. The list will be updated once the project 2 start comes closer.

Other labs are possible as well and you're encouraged to reach out to any lab of your choice (even outside EPFL), as mentioned above.

Important Logistics: You can only sign up for this option if the *professor* of that lab agrees to host your group of 3 students, and will confirm this by filling the mandatory registration form for labs, by November 12 (ideally).

Final Submission of Your Project

Submission URL for the final project: `http://mlcourse.epfl.ch`

Your final submission to the online system (a standard system as used for scientific conferences) must consist of the following:

- **Report:** Your 4 page report as `.pdf`. An additional appendix is allowed. References and acknowledgements are also allowed to be put on extra pages. Please choose a meaningful title to your report. Be aware the appendix might not be graded by the reviewers necessarily, but can help to support your main claims.
- **Code:** The complete executable and documented Python code, as a link to your team's github repository. The link should point to your final commit, in the format `https://github.com/your-username/your-proj-name/tree/the-last-commit-hash` (you can get it by pressing 'y' on the github webpage of your repository). Rules for the code part (see also the guidelines further below)
  - *Reproducibility:* Please provide well-documented and reproducible code. This can be in form of Python notebooks or other form. Include a clear `ReadMe` file explaining your code organization, and how to reproduce your setup. Include pre-trained models when possible, as well as additional code to produce them, including all data preparation, feature generation as well as cross-validation steps that you have used. Include all needed data whenever possible. Github has a 100MB file size limit; for larger data, clearly explain where to obtain it (e.g., via a timestamped external file hosting service), and ideally automatic download scripts.
  - *Documentation:* Your ML system must be clearly described in your PDF report and also well-documented in the code itself. The documentation must also include all data preparation, feature generation as well as cross-validation steps that you have used.
  - *External libraries, models and datasets* are allowed, as long as accurately cited and documented.
- **AI usage:** The use of AI tools is allowed, but you are required to describe it in the appendix of your report. Please explicitly state how you used LLMs in the research work (which is valuable information for everyone also). Note that ultimately the paper's authors are responsible for the contents of their submissions. See related scientific venues also, such as ICLR's recommendation.

Don't forget to give a meaningful title to your project, both in the form and on the PDF. The title should not be 'Project 2' but a descriptive title of your approach. Please provide all author names with their EPFL email in the form. You can update all parts of your submission anytime until the deadline.

Ethical Risk Assessment in the Written Report

You should evaluate the ethical risks of your project at least once, preferably in the early phases. To assist in this process, we suggest that you use the Digital Ethics Canvas, a risk assessment grid with a series of questions that guide you in the analysis of software-specific ethical risks.

In your report, you **must include an "Ethical risks" section** (200 to 400 words max., does not count in the 4-page limit), which counts for **part of the grade of the project report** and should consist of one or two short paragraphs. You may choose one of the two options based on your unique project topic:

Situation 1: You have identified at least one ethical risk

Focus on *one* of the ethical risks you identified and answer the following questions:

- Describe the risk

  - Who is impacted (which type of stakeholder)?
  - What is the negative impact?
  - How significant is the risk, both in terms of severity and likelihood of occurrence?
- How did you evaluate this risk (research you did, metrics you measured)?
- How have you taken this risk into account in your project?
  - If you were able to take it into account, what did you change?
  - If you were not able to take it into account, what were the barriers?

Situation 2: You have not identified any ethical risk

Please describe the process you followed to rule out the presence of ethical risks:

- Which types of stakeholders did you consider when evaluating the ethical risks of your project? You should mention at least 2 categories, including some indirect stakeholders and/or the environment, and provide a description of their roles.
- How did you rule out the different ethical risks for these stakeholders (research you made, metrics you measured)?

You are welcome to attach your Digital Ethics Canvas as an Appendix to the report to help illustrate your risk analysis process, but it is not required.

Appendix

Best Machine Learning practices

We recommend using the pytorch project template proposed by LauzHack. While it is entirely optional, it will help you learn practical ML skills and tools. This includes configuration files with Hydra, automatic code checks with pre-commit, and a clear repository organization. To learn more about pytorch concepts, you may also be interested in LauzHack's events and the deep-learning-bootcamp, which covers many of the fundamental principles and implementation details that are not part of this class.

Grading Criteria and Some Advice

We are aware that not all projects have the same difficulty. We will take the difficulty of the question into account when grading, so none of the choices will have any disadvantage for you.

- **Code.** In Python. You are allowed to use external libraries, as long as properly cited, documented and referenced. Your project submission will be graded by two assistants independently. As some additional advice for the code part,
  - Double-check that the archive you submit actually contains everything needed to run your solution.
  - Make sure any additional installation needed to run your solution is fully described in the accompanying `ReadMe` file, including version numbers if necessary.
  - Try to follow your own instructions to get your code running at least once.
  - Try to make your code as readable as possible. Reduce copy-pasted code to a minimum, use helper functions and clearly name your files, functions and variables.
- **Written Report.** You will write a maximum 4 page PDF report on your findings, using LaTeX. We will grade you on the scientific contribution you made, that is on the improvement achieved over the standard baseline methods, as well as the rigorous and fair measuring of the claimed improvements. The criteria are

  - **Solid comparison baselines supporting your claims**
    Quantify the benefits of your method by providing clear quality measurements of the most important aspects and additions you chose for your model. Start with a very basic baseline, and demonstrate what improvements your contributions yield.
  - **Reproducibility**
    Your classmates should be able to reproduce your results based on your report only. Describe what preprocessing is required, what hyper-parameter values you selected and why, and clearly describe the overall pipeline you used.
  - **Scientific novelty and creativity**
    You will likely be using more than the standard methods we saw in the first half of the course. To communicate that your methods work and that you understand them, you should make sure that your report makes clear the following points.
    - What *specific* problem your method is intended to solve.
      By specific, we do not mean "image classification" but what specific issue with your current model are you trying to improve with this method.
    - Why is this an important problem? Why are you solving this one instead of something else?
    - How is your method helping?
    - What are the results of your method? Compare the error before and after.
  - **Ethics component**
    Situation 1:
    - The risk should be described specifically in the context of the project and should include elements on the affected stakeholder(s), the type and importance of the potential impact(s).
    - The description of the risk evaluation process should include references to existing sources documenting the risk or refer to metrics used to evaluate the risk in the context of the project.

  - The text should describe how the results of the risk analysis have been considered in the project, either by presenting specific changes made to the project or by detailing specific barriers to mitigation.

  Situation 2:

  - At least 2 different categories of project stakeholders should be presented with a description of their role, and should include some indirect stakeholder(s) and/or the environment.
  - The description of the risk evaluation process should include references to existing sources ruling out the risks or refer to metrics used to rule out the risks in the context of the project.

- **Writeup quality**

  Some advice:

  - Try to convey a clear story giving the most relevant aspects of your approach, in a reproducible way. Learning what has not worked can additionally help the reader (and help them better understand *why* you have made the many choices you did), but focus on what is most relevant for your final solution.
  - Before the submission, have an external person proofread your report. It is easy to write a sentence that makes perfect sense to you since you wrote it but is actually hard to parse. Use a spell-checker.
  - Plots are great way to share information that might be hard to convey by writing. Make sure that your plots are understandable, have labels for axes, a title, correct axes limits, add a description of what your plot is about and what can be learned from it.

As usual, your code and report will be automatically checked for plagiarism.

Guidelines for Machine Learning Projects

Now that you have implemented few basic methods, you should use this toolbox on the dataset. Here are a few things that you might want to try.

Exploratory data analysis You should learn about your dataset - figure out which features are continuous, which ones are categorical, check if there are obvious relationships between the features, take a look at the distribution of each feature, and so on. Check https://en.wikipedia.org/wiki/Exploratory_data_analysis.

Feature processing Cleaning your dataset by removing useless features and values, combining others, finding better representations of the features to feed your model, scaling the features, and so on. Check this article on feature engineering: http://machinelearningmastery.com/discover-feature-engineering-how-to-engineer-features-and-how-to-get-good-at-it/.

Determining whether a method overfits or underfits You should be able to diagnose the whether your model is over- or underfitting the data and take actions to fix the problems with your model. Recommended reading: *Advice on applying machine learning methods* by Andrew Ng: http://cs229.stanford.edu/materials/ML-advice.pdf.

Applying methods and visualizing Beyond simply applying the models we have seen, it helps to try to understand what the ML model is doing. Try to find out which datapoints are wrongly classified and, if possible, why this is the case. Then use this information to improve your model. Check Peter Domingo's *Useful things to know about machine learning*: http://homes.cs.washington.edu/~pedrod/papers/cacm12.pdf

Accurately estimate how well your method is doing By applying cross-validation and estimating the test error.

Report Guidelines

In addition to finding a good model for the data, you will need to explain your methodology in a report.

Clearly describe your used methods, state your conclusions and argue that the results you obtained make (or do not make) sense, and the reasons behind it. Keep the report short and to the point, with a strict limit of 4 pages. References and a potential ethics statement are allowed to be put on extra pages.

To get started more easily with writing the report, we provide you a LaTeX template here

github.com/epfml/ML_course/tree/main/projects/project1/latex-example-paper

The file also contains some more helpful information on how to write a scientific report or paper. We will also help you learn it during the exercise session and office hours if you ask us.

For more guidelines on what makes a good report, see the grading criteria above. In particular, don't forget to take care about

- *Reproducibility:* Not only in the code, but also in the report, do include complete details about each algorithm you tried, e.g. what lambda values you used for ridge regression? How exactly did you do that feature transformation? how many folds did you use for cross-validation? etc...
- *Baselines:* Give clear experimental evidence: When you added this new combined feature, or changed the regularization, by how much did that increase or decrease the test error? It is crucial to always report such obtained differences in the evaluation metrics, and to include several properly implemented baseline algorithms as a comparison to your approach.

Some additional resources on LaTeX:

- https://github.com/VoLuong/Begin-Latex-in-minutes - getting started with LaTeX
- http://www.maths.tcd.ie/~dwilkins/LaTeXPrimer/ - tutorial on LaTeX
- https://wch.github.io/latexsheet/ - cheat sheet collecting most of all useful commands in LaTeX
- http://en.wikibooks.org/wiki/LaTeX - detailed tutorial on LaTeX

Producing figures for LaTeX in Python

There are some good visualization tools in Python. "matplotlib" is probably the single most used Python package for 2D-graphics. The relevant tutorials are as follow:

- Matplotlib tutorial: https://github.com/rougier/matplotlib-tutorial
- Matplotlib tutorial other useful Python data visualization libraries: http://jakevdp.github.io/mpl_tutorial/